→ International organization of standardization
→ Open system Interconnection

ISO/OSI : [Divide & Conquer, Encapsulation, Abstraction, Testing] → Adv. of Layering

→ Physical Layer : → Bit sync | speed of sender & Receiver
→ Bit rate Control
→ Topologies → Encoding [Manchester etc]
→ Transmission mode [full or simple or half duplex].

Hub, Repeaters, Modem, Cables.

[for Manchester Baud Rate = 2 * Bit Rate]

→ Data Link Layer : → HDLC [Bit stream protocol]
→ Host to Host delivery → Uses MAC address [Physical Address]
→ Logical Control (Unit)
→ Error Control & detection [CRC, checksum]
→ Flow Control [S&W, GBN, SR]
→ Media Access Control [CSMA/CD, Aloha]
→ Data in frames.
→ Framing
→ Physical Address → Error Control

Switch, Bridges

| M | x | B_0 | S_A | S_A | N_A | M_H |

MAC Address

Framing

→ fixed length
[Internal fragmentation ↑]
[Rarely used]
[Bit / stuffing more req.]

→ Variable length
[segmentation, start delimiter, end delimiter (ED) required or put frame length in field Also].

for ED

Character Del. / stuffing
↓
'/0' → to Represent End of msg.

Bit stuffing
↓
ED 01111.
Msg 011111 → 0111101

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Que Bitstuffing- ED - 0111
M - 01110110 ⇒ 01101011100 Op.

→ Network Layer :
- → Host to host Connectivity
- → Logical Addressing [IPv4, IPv6]
- → Routing
- → Switching [packet, circuit, datagram]
- ** → data in packet.
- → Congestion Control [Not in GATE]
- → fragmentation

[m | a | S_0 | S_A | S_D]
S_A ↓ IP source, S_D ↓ IP

Switching
- Circuit (FDM, TDM) Switching
  - → Work on physical Layer
  - → dedicated path & resources
  - → Continuous flow
  - → Out of order not possible
  - → No Headers
  - → Delay less + Efficiency less
- Message switching
- Packet switching
  - ** → Works on N/W & Data LL
  - → Async transfer mode
    - Virtual Ckt
      - → At DLL
      - → Connection oriented
      - → No out of order
      - → Less overhead
      - → packet loss less
      - → Cost ↑
      - → phone call eg.
    - Datagram
      - → At N/W
      - → Connection less
      - → out of order
      - → More overhead
      - → packet loss More
      - → data call eg. [eg Fb, Whatsapp call]

Total time = setup + T_t + T_{pt} + teardown

Eff. BW = $\frac{BW}{\#slots}$

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→ Session Layer : → Authentication & Authorization
→ Checkpoints
→ Synchronization
→ session establishment, maintanance & Terminatn
→ Dialog Controller
↓
[Allow two sys to start Commn in Half or full duplex].

→ Presentation Layer :
→ Character translation
→ Encryption & Decryption
→ Compression

→ Application Layer :
[m] → message

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Circuit switching	Packet switching (Lp. points)
✓ Congestion can occurs during connection establishment.	→ Store & forward [uses Routing Algo]
	→ More efficient
	→ No dedicated path
	→ pipelining used
✓ No forward & store.	→ More delay
	→ Congestion can occur during transfer time.

Total time = $n(T_t) + n\,T_p$

n ⇓ No of Hops.

→ **Transport Layer:**

→ Heart of OSI
→ End to end (process to process) & service point Addressing
→ Port No.
→ segmentation & Assembly
→ Flow Control & Error Control ↓ Make sure Packet Received by Correct process
→ [SR, GBN etc]. [CRC, Checksum] ↓ H/W ↓ S/W

[m | x | 80]
m → Source port
x → destination port

→ Multiplexing & Demultiplexing
→ segmentation
→ Congestion control [open - Prevent, close - Remove]
→ data integrity

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Flow Control

Imp [1KB - 1024 B
1kbps - 10^6 bps] ***

⇒ Bcoz BW is measured in term of freq. & freq in term of 10. power

⇒ Delays [Transmission time (T_t), T_p, Queing delay, Propogation delay.].

Stop & Wait :

✓ → piggybacking - [Ack with data]

So. T_t / $T_t + T_{t_{Ack}} + T_{p_{data}} + T_{p_{ACK}} + T_{queue} + T_{processing}$

($\neq 0$). ($\neq 0$) (time ≥ 0) (time $\neq 0$).

$\eta = \frac{T_t}{T_t + 2T_p}$ ⇒ Assumed $T_{t_{Ack}} \approx 0$.

Imp: In piggybacking, $T_{t_{Ack}} = T_{t\,Data}$

✓ → Throughput / Effective Bandwidth = $\eta * BW$

✓ Problems: ① Data packet lost: data lost, No Ack. So, sender will wait for Ack & Receiver for new packet.
⇒ Deadlock.
→ Add Timer Out. S&W + TO ⇒ S&W ARQ.

② Ack is lost: → It will Leads to duplicate data.
Receiver will discard. So, Add SEQ No
→ to each packet.

③ Delayed Ack: → Ack of first will delay & Reached after second.
→ So, Add Ack No with each Ack.

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Imp → Retransmission - only for lost packets

→ Seq. No = 2 [$W_s = 1, W_R = 1$]

→ Suitable for LAN not for WAN [d↑, η↓]

→ Not suitable for thick pipe [BW↑, η↓]

↓ High capacity

→ Buffer size [1+1] (Sender ↑, Receiver ↓)

→ BW is low → CPU is low

pipelining:

T_t sec —— 1 packet

1 —— $\frac{1}{T_t}$

$t_t + 2*T_p$ —— $\frac{T_t + 2*T_p}{T_t} = (1 + 2a)$

So, for η = 100 %, # packet in $T_t + 2*T_p$ should be $T_t + 2T_p \Rightarrow (1+2a)$

Can be achieved by pipelining.

→ Sliding window protocol: → GB N.

→ Selective Repeat [$W_s > 1, W_R > 1$]

W_s (Window size) $= 1 + 2a$

seq. $= 1 + 2a$

bits required for seq. $= \lceil \log_2 (1+2a) \rceil$

Note: $W_s = \min\{1 + 2a, 2^N\}$ → N = # bits for sequence No.

Imp.

Que.

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Imp: $(W_s + W_R) \leq$ AtLeast sequence No → valid for all packet Protocol.

[to eliminate duplicate packet issue]

⇒ GB N:

→ Sender window size = N
→ Receiver window size = 1.

⇒ # seq. Nos = (N+1)

$$\eta = \frac{N}{1+2a}$$

→ Let sending and packet 2 Lost. and 3, 4, 5 already reached. Receiver will discard [3,4,5] bcoz waiting for 2. # Called Sanity Discard.

GB3, Every 5th lost.

0 1 2 3 (4) 5 6 7 (8) 9 10

~~0 1 2 3 4 5 6 4 5 6 7 8 9 10 8 9 10~~

Bus

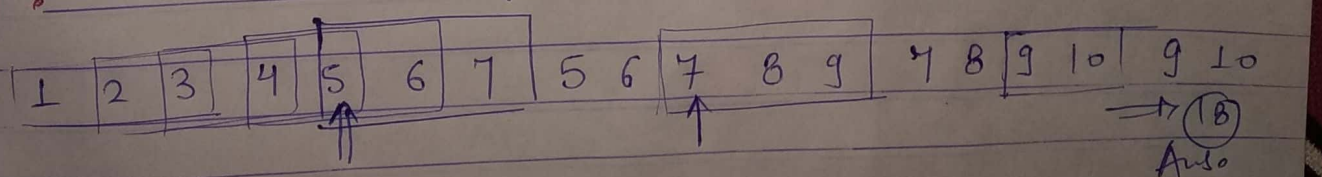
Re-
→ Transmission of whole window.

→ Cummulative Ack [Single Ack for Multiple packets].
↳ [Adv.: Less Traffic, Dis. - Less Reliable]

→ Buffer $[W_s + W_R = N+1]$.

→ BW : Very High → CPU-Medium
→ Implementation : Medium

Col^n: ⇒ discard packet if Corrupted.



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So that Acknowledged packet will be send earlier. No need of Ack TO. fp.

⇒ Selective Repeat. fp → send NAck if packet is corrupted.

- Sender $W_S > 1$
- $W_R > 1$ → BW is Medium.
- Buffer [$W_S + W_R$ → N+N].
- Independent Acks.
- Only lost packet will be re-transmitted.

fp → sorting might Required. [packet might not be in order]

$$\eta = \frac{N}{1+2a}$$

→ Re-transmission is easy. [only lost packet].

Que. $W_S = 3$, 10 packets & every 5th lost.

1 2 3 4 5 6 7 8 9 10

⇒ 1 2 3 4 5 5 6 7 8 9 9 10 → (12) Ans.

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Error detection & Control

⇒ Min^m no of redundant bit req to correct one bit error -

$m + r = n$

- m → Msg.
- r → Redundant
- n → Codeword Length

$$2^m(n+1) \leq 2^n \quad **$$

⇒ $2^m(m+r+1) \leq 2^{m+r}$ ⇒ $(m+r+1) \leq 2^r$ **

↓ Necessary Cond^n not sufficient.

Hamming Code: → error correction code.

⇒ Min^m Hamm^n Distance to detect 'i' bit error $\geq i+1$
to correct 't' " $\geq t+2$ **

↳ It uses extra parity bit.

General Algo:

1. Write the bit position starting from 1 in binary form.
2. All the bit position that are power of 2 are marked as parity bits [1, 2, 4, 8 etc]
3. All other bits are marked as data bit.
4. Each data bit is included in Unique set.
   - ⇒ Parity bit 1 will cover all whose least sig is 1.
     eg. [1, 3, 5, 7, 9 etc]
   - ⇒ parity bit 2 will cover all the bits positions whose binary rep includes a 1 in the second position from

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from Least significant bit [2, 3, 6, 7 etc]
↓
Continue.

5. Now, according to odd or even parity mechanism, update parity bit position.

✓ **Checksum:** → Used At N/W & Transport Layer.
→ Add data segments. If carry then Add that Also.
→ Send compliment of ~~all~~ Addition.
↳ At Receiver, Add that checksum. If o/p = zero. Msg is correct without any error msg.

```
  10001001
  11110001
  ---------
  01111010
         1
  ---------
  01111011
```

↓
Add
↳ Compliment of final Addition.
↳ Send with Msg.

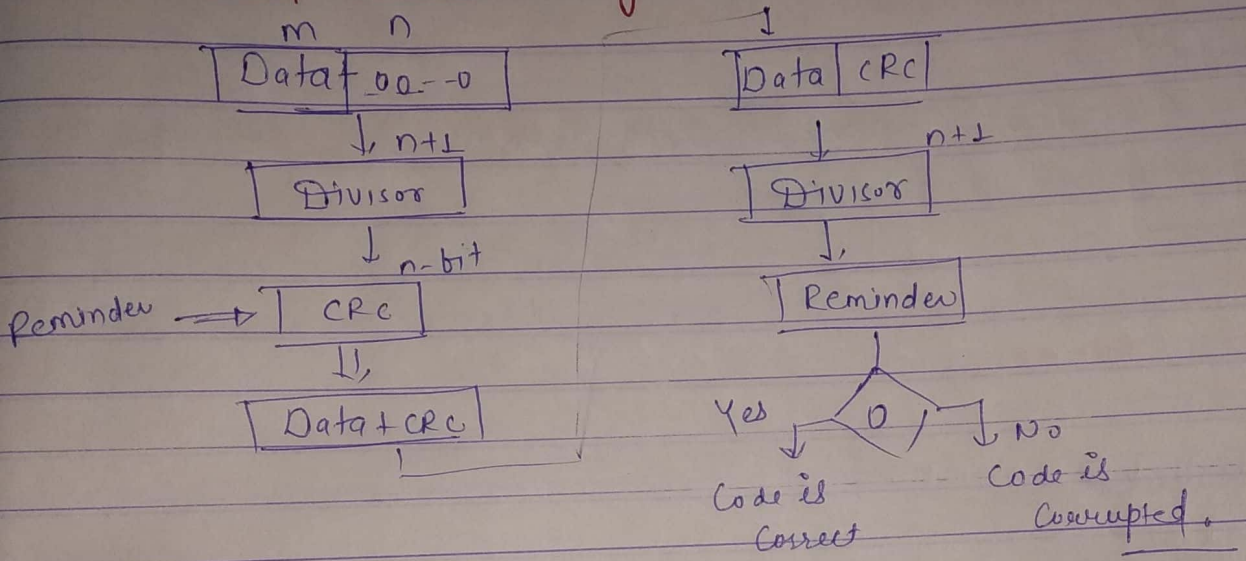
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⊕ · EX-OR ·

⇑

[Mod 2 Sum]

✓ CRC (Cyclic Redundancy Code) : ⟹ We do



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Medium Access Control

⇒ Random Access Protocol:

① **Pure Aloha:** → Wait for Ack

→ $\eta = G * e^{-2G}$ $\quad G = 1/2 \Rightarrow \eta = 0.184$

→ Vulnerability time $= 2t_p$

→ If Ack doesn't Come, then wait for sometime back-off time (T_b)

② **Slotted Aloha:** → Wait for Ack

→ $\eta = G * e^{-G} \Rightarrow \eta = 0.368$

→ Vulnerable time $= t_p$.

→ Every station tries at Random moment, to resend data. so collision is Less.

③ **CSMA:** → check for Medium, if idle then send the data.

- → 1-persistent: Contin check, if idle 100% Resend.
- → 0-persistent: Based on station priority, wait for slot.
- → Non-persistent: check After Random amount of time
- → p-persistent: Send with p-probability if idle.

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A*

→ Also called Ethernet.

④ CSMA/CD: → No Ack

→ Not suitable for long distance. So, used for LAN.

→ packet size ↑, ⇒ η = ↑. But we have to limit the size of packet. otherwise that will cont^n transfer for single station

A**

$$\eta = \frac{1}{1 + 6.44a} > \frac{1}{C * 2 * t_p + t_p + t_t}$$

No of collision

Imp:

→ $t_t \geq 2 * t_p$ ⇒ To detect collision

→ Back-off Algorithm [Also called Binary exponential Algo].

→ Capture effect: station who wins once, keeps on winning.

$k = [0, 2^n - 1]$.

waiting time = k * slot

→ Collision Resolution

→ only for 2 stations

→ every time collision probability will decrease.

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(5) CSMA/CA:

→ Interframe space: Time duration for each station. In that time only station can send data.

→ Ack send

→ Contention window

⇒ Control Access Protocol:

(1) Reservation (TDM): → Each slot to each station.

→ $N * BW_s = \text{Total BW}$

→ $\eta = \frac{t_t}{t_t + t_p}$

(2) Polling:

→ $\eta = \frac{t_t}{t_{proc} + t_t + t_p}$

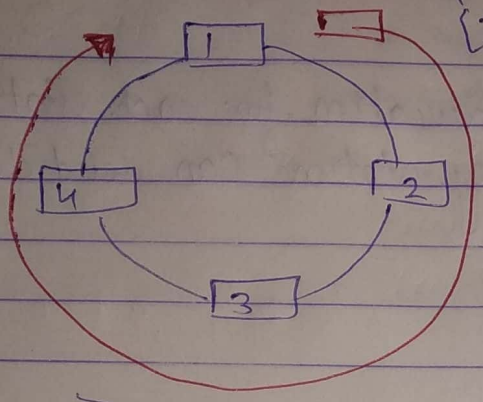
→ Starvation may possible.

(3) Token's Ring: → Delayed token Retrain: $\frac{1}{1+\frac{(N+1)}{N}a} = \eta$ ⇒ More Reliable

→ Faulty token Retransmission: $\frac{1}{1+\frac{a}{N}} = \eta$ ⇒ Less Reliable.

By default method.

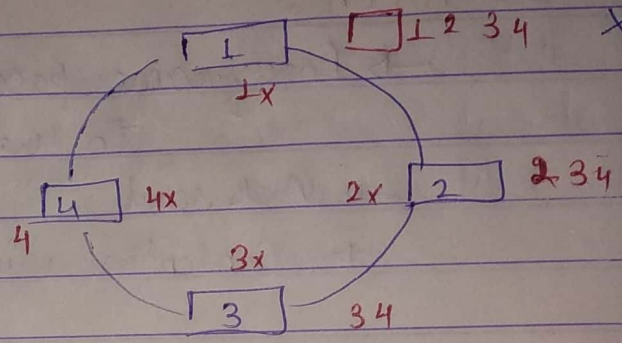
Scanned by CamScanner



→ delayed token [Token held till end cycle].

$$\eta = \frac{N * T_t}{T_p + N * (T_t + T_p)}$$

⇒ Early token Release. Each station will Release its own token.



$$\eta = \frac{N * T_t}{T_p + N * T_t}$$

**

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LAN Technologies

① Ethernet (IEEE 802.3)

→ Work on physical and data link Layer.
→ CSMA/CD - 1-persistent
→ No Ack
→ Topology - Bus
→ Data Rate: 100 Mbit/s → fast
1 Gbs/s → Gigabit
10 Gigabit
→ Manchester Encoding [Baud Rate = 2 * Bit Rate]

hp.
ep
→ AL - Message
TL - segment
NL - Datagram
DLL - Frame
PL - Single Protocol data unit (SPDU)

→ Network Node → DTE (Terminal Equipment: Signal to data & Vice Versa)
↳ DCE [Comm^{n} Equipment: switch, Routers etc.]

(7-byte) Preamble	(1B) SFD	(6B) Desti Add (MAC)	(6B) Source Add (MAC)	(2B) Length	(46-1500 B) Data (Payload)	(4B) CRC

Set 10101011
Before actual data stream.

(46 46-1500 Byte ⇓ Total data Size

Generating using Source ADD. + Destination ADD. + Data + Length

Total size of frame = 64 to 1518 Bytes

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→ station send jamming signal if collision is detected.

② Token Ring (802.5):

↓
Check page notes [Images].

- → Topology - Ring
- → Access Control - Token passing access method.
- → Uni-direction (simplex)
- → kind of Ack - piggybacking
- → Data Rate : 4 Mbps, 16 Mbps
- → Encoding Technique : DME

→ Problems At Sender:

① Orphan packet: Sender went down, no one will accept data.
- ↳ Used monitor bit. [m = 0, initially, if Received then change bit to 1].

② Stray packet: packet is corrupted. Sender failed to recognize.
- ↳ CRC used. If corrupt, discard.

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of Wi-fi As

⇒ Problem at destination:

⇒ Packet Problem: → Every station has to hold the token ~~problem~~ and then send data.

↳ Starvation may occur.
↳ define Maxm Token holding time.

Que. Maxm data that can be send =
Token Holding time * BandWidth

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RTS- Request To send
CTS- Clear To send

(wifi – wireless fidelity)

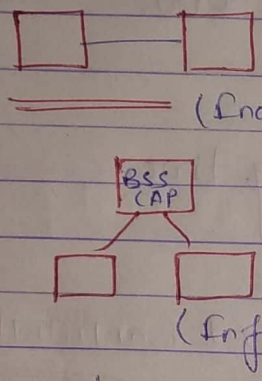
(3) Wireless LAN (IEEE 802.11):

↳ Covers physical (Level 1) and data link (Level 2) layer.

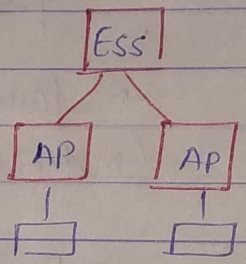
↳ BSS (Basic service set), ESS (Extended Service set)

↓

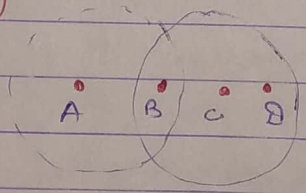
BSS



ESS



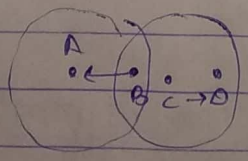
↳ Problems: ① Hidden Substation Problem:



⇒ A & C both send data to B.
A & C both are out of Raye
So, Collision detection not possible.

↳ To overcome – RTS/CTS Handshaking in addition CSMA/CA.

② Exposed Node problem:



⇒ B is sending. C can't send as it thought B is sending. But B is sending to A & C want to send to D. Both will not affect but can't send.

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We Address Above problms of Wi-fi As Multiple Access with Collision Avoidance.

Wifi - Continue.....

→ RTS | CTS Algo:

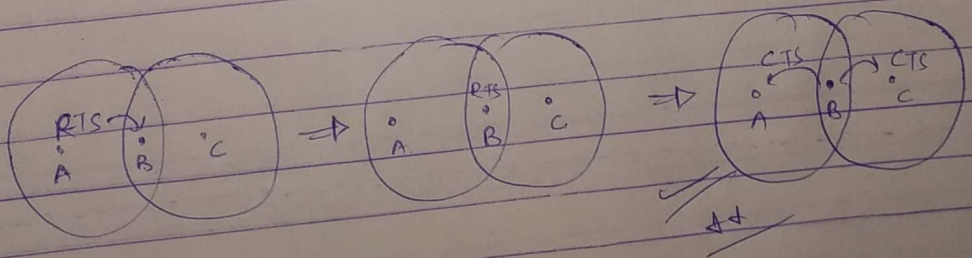
→ The sender and Receiver will exchange frames with each other before transmitting the data.

→ This informs all nearby nodes that a transmission is about to begin.

Sender will send - RTS
Receiver will reply - CTS

→ Any nodes that receive CTS, node will know that it is close to Receiver, therefore can't transmit a frame.

→ Any node that Receive ~~RTS~~ CTS but not ~~CTS~~ RTS frame knows that it is not close to the receiver to interfere with it. So, free to transmit.



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Transport Layer

⇒ Congestion Control:

Open Loop
(Prevent the Congestion)

During Comm^{n} setup, the transmission rate is determined.

"Once the connection is set-up, packets are sent and processed by the system without feedback info^{n} from either the N/w or destination."

- ↳ policing [Leaky Bucket & Token Bucket] (Traffic shaping)
- ↳ call Admission Control: All Resources got fixed.
  [QoS parameter will define arrival rate / Traffic rate etc in a N/w]

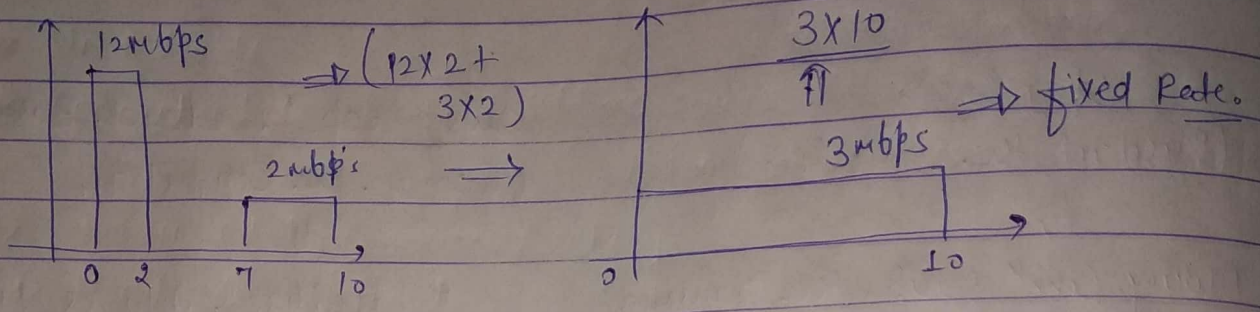
Close Loop / Feedback Based Control
(Remove the Congestion.

"feedback info is provided by N/w or destination"

- ↳ Senders can change forward speed dynamically.
- ↳ Back pass : Node to Node
- ↳ Choke packet : Direct to sender
- ↳ Explicit signal : To sender with msg.
- ↳ Implicit signal : source guess for congestion if no Ack reached.

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① Leaky Bucket:

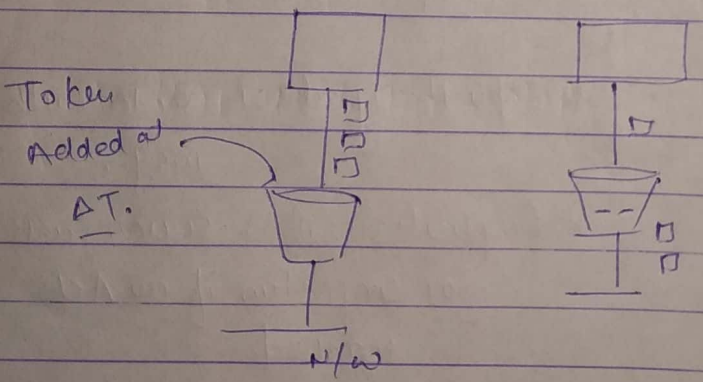


↳ If data is comming to much, then bucket itself may overflow.
[Resolved by Token Bucket]

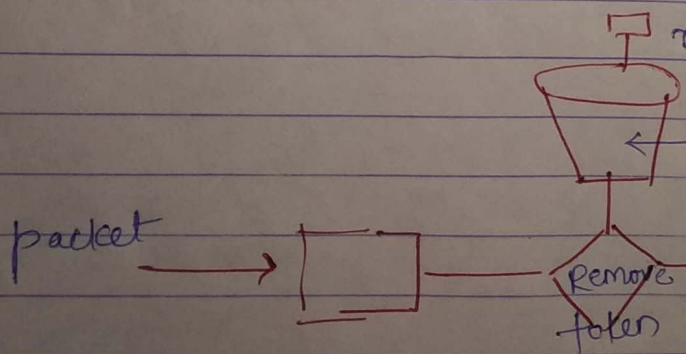
Ip. ↳ Uniform outlet is insured.

② Token Bucket:

↳ Uniform outlet is ~~not~~ insured.



⇒ The no of packet that can be added will equal to the # tokens.



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So, In a time interval, 't' the data that can be transferred -

$$r * t + b = \text{data}$$

↓

[Max^m o/p Rate * t] b/p.

Que : Token Bucket capacity = 250 KB (b)
Token arrive rate = 2 MB/sec. (r)
If Max^m o/p Rate = 25 MB/sec. (data)
then Burst time = ? (t = ?)

$$250 + 2 * t = 25 * t \Rightarrow t = 25.$$

$$\Rightarrow 250 = (25 - 2)t$$

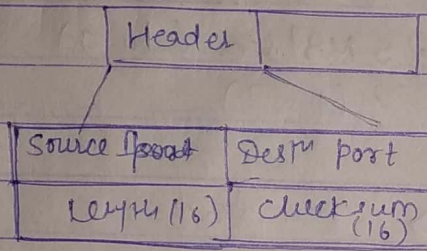
$$\Rightarrow \boxed{t = \frac{250}{23}} \text{ Ans.}$$

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⇒ UDP Protocol:

↳ Un-Reliable
↳ Connectionless [Eg. Realtime service like Gaming, Voice or Video Comm^n, Live Conferences & Multimedia App.]
↳ No mandatory error checking
↳ Header is fixed.

8 bits



⇒ UDP Header + Pseudo Header of info from IP Header + Data.

Qp. ↳ support packet switching & better for multicast or BroadCast.

Qp. ↳ UDP used for - NTP, DNS, BOOTP, DHCP, NNP, TRTP, RTSP, RIP, OSPF

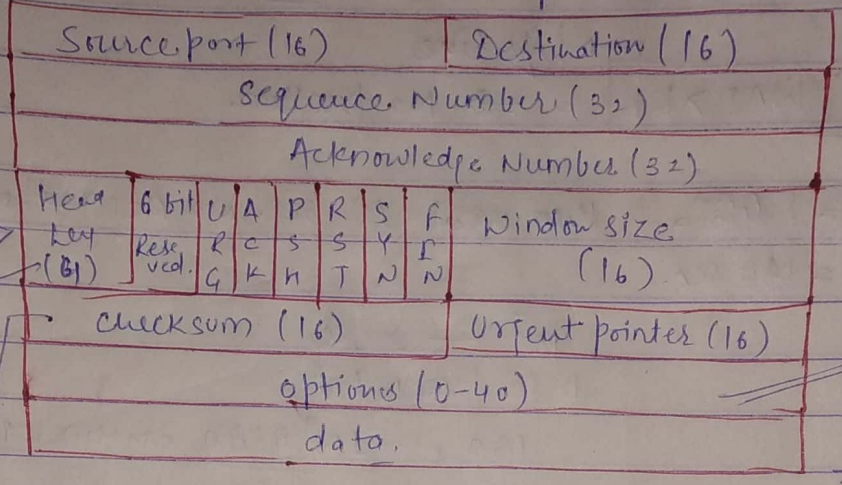
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TCP

Socket = IP + Port
(32) (16) ⇒ Uniquely identify a process.

HTTP - 80
FTP - 21
Telnet - 23
SMTP - 25

Header:



(13×4 = 60, 5×4 = 20) ⇒ (20 to 60) Bytes.

TCP Header + data from IP Header

Timestamp, window size extension, padding

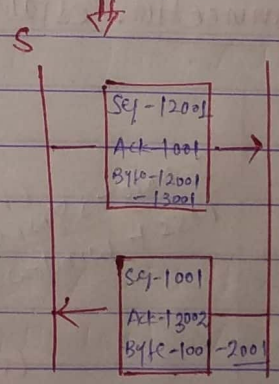
⇒ Uses piggybacking

All well known port - 0 to $2^{10}-1$
Reserved - 2^{10} to 49151
General public - 49152 to 65535.

⇒ (Byte stream)
⇒ Stream Oriented (In segments)
⇒ Connection Oriented (3-way Handshaking)
⇒ Full duplex services

Rfp.

⇒ Sequence No not start with zero [Generally don't use] we use Random initial sequence Number.



⇒ Wrap up time:

Let we have seq NO = 2^{32}
So, 4G data can be send.
Suppose data size is more than 4G. Then use same sequence no which are already been used.

eg. BW = 1MBPS
2^{32} sequence No
Then
WAT = ?

1Mb — 1 sec
10^6 — 1 sec.
1 — $\frac{1}{10^6}$

$2^{32} \Rightarrow \frac{2^{32}}{10^6}$ sec.

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→ If BW is high. then WAT is low.

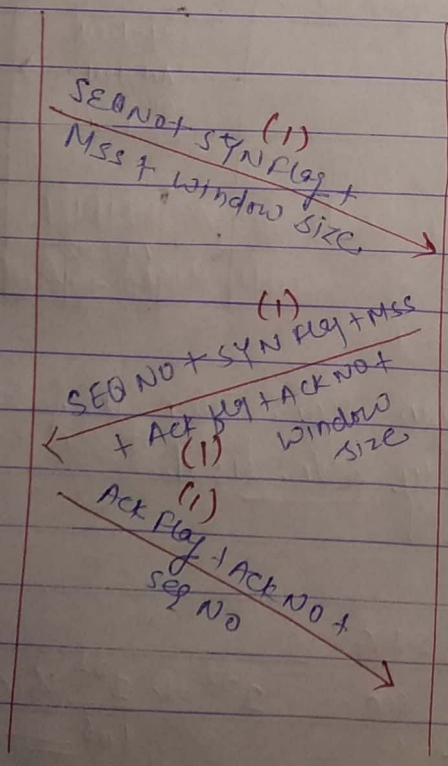
↳ if [WAT > LT] → Life line ⇒ No problem.

Que. BW = 1 GBPS then what is seq. No Reqred Min^m. ?
LT = 180 sec.

10^9 — 1 sec.
1 — 10^9
180 — $10^9 * 180$ ⇒ Min seq. No.

Bits = $\lceil \log_2 (10^9 * 180) \rceil$

⇒ Connection establishment: (SYN, SYN + Ack, Ack)

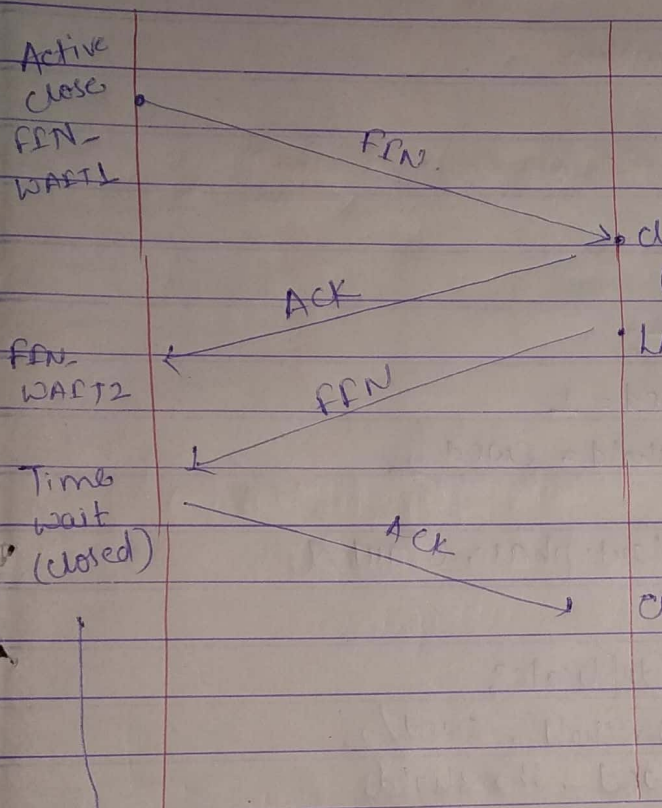


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→ Connection Termination:

FIN: Gracefully terminate the Connection.
→ No data loss.

RST: Abrupt termination
→ Data discarded.



↳ The TIME_WAIT lets the client resend the final ACK in case the ACK is lost.

→ Persistant Timer:

Assumed from B, Ack is lost. so, A will think B don't want more data and B will think A don't have more data. So, Deadlock.

↳ so, use persistant Timer. A will send 1 Byte of data just to check B. is awake or not.

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⇒ TCP Congestion Control:

→ Start phase: cwnd = $2^1, 2^2$ ...

→ Congestion Avoidance: cwnd = cwnd + 1

→ Congestion detection:

⇒ If timer out:

cwnd = 1

Threshold = cwnd/2

and start phase started

⇒ 3 Ack ~~last~~/duplicate:

threshold = cwnd/2

Cwnd = threshold

and

Congestion Avoidance phase

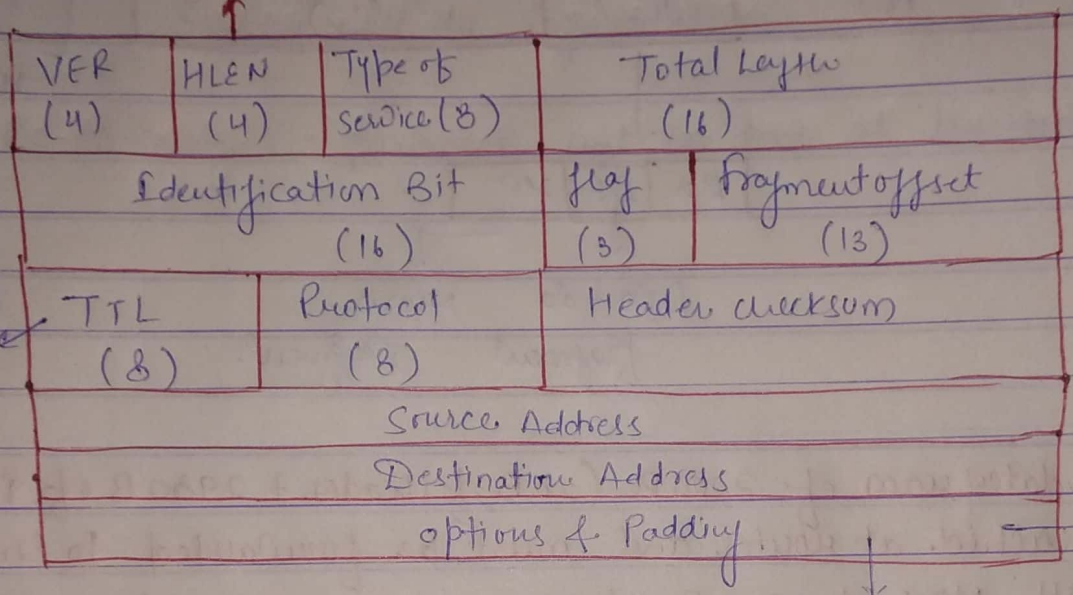
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IPv4

⇒ Packet stream protocol

(20-60)

Protocol: UDP, TCP, ICMP, IGMP etc.



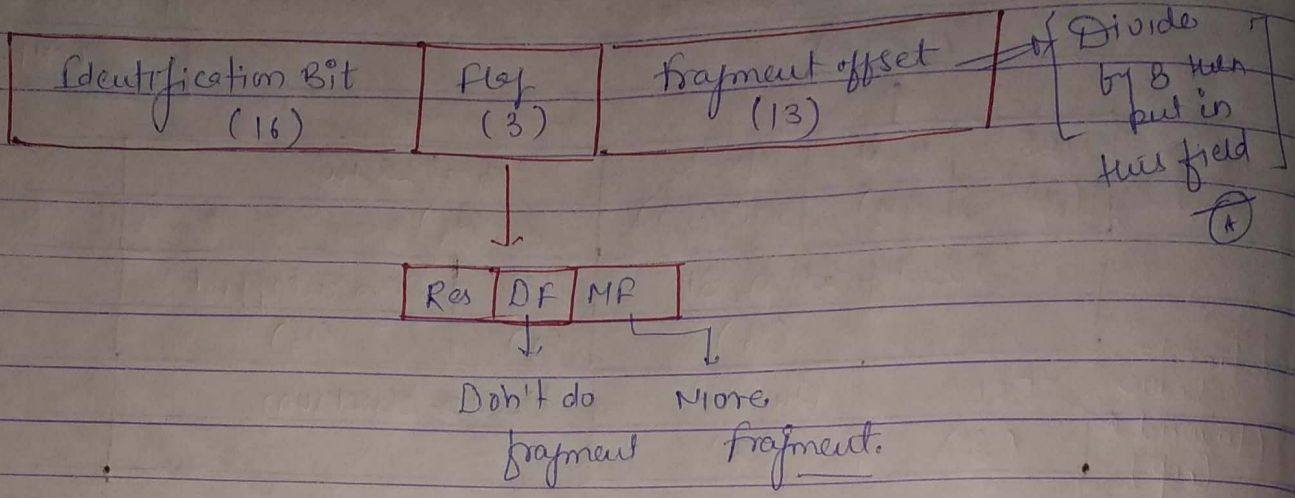
↳ 32-bit
↳ Connectionless
↳ Best effort delivery model.
↳ Under RFC 791
↳ Uses five classes A, B, C, D, E

Sup → Let HLEN = 30. ⇒ So, How will will store in HLEN field?

⌈30+2⌉ + padding ⇒ 32.

So, Stor HLEN = 8.

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Ques. A datagram of 3000 B (20 B of Header + 2980 B of IP Payload) reached at router and must be forwarded to link with MTU of 500 B. How many fragment will be generated. and Also write MF offset etc.

MTU = 500 B [20 B + 480 B]

→ # fragments:

$$\left\lceil \frac{2980}{480} \right\rceil = 7$$

P_1	P_2	P_3	P_4	P_5	P_6	P_7
480 +	480 +	480 +	480 +	480 +	480 +	100
20	20	20	20	20	20	+ 20

⇒ **offset**: How many bytes of data ahead.

P_1 → 0

P_2 → (480) / 8 = 60

P_3 → (480 + 480) / 8 = 120

P_4 → ____

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→ Fragmentation will be done at routers not at source.
Source will do Segmentation.

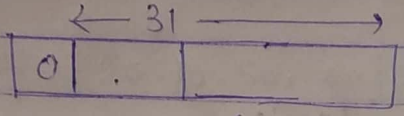
→ Re-Assembly of fragments will be done at the destination.

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Network Layer

⇒ Classful:

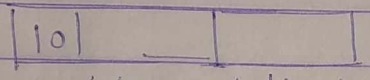
→ Class A: (1 to 126)



Host = 24 bits
NID = 8 bits

N/W = 2^7
I/P = 2^{24} IPs / N/W
Hosts = $2^{24} - 2$.

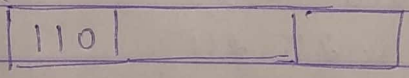
→ Class B: (128 to 191)



Host = 16 bits
NID = 16 bits

N/W = 2^{14}
I/P = 2^{16} IPs / N/W
Hosts = $2^{16} - 2$

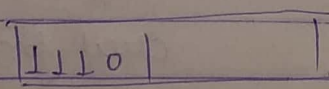
→ Class C: (192 to 223)



Host = 8 bit
NID = 24 bit

N/W = 2^{21}
IPs = 2^8 IPs / N/W
Hosts = $2^8 - 2$ / N/W.

→ Class D:



→ Used for multicasting.

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IPs = 2^n
Hosts = $2^n - 2$

→ Class E:

→ Reserved Category.

Broadcasting

Limited Bc

↳ To all host of a Network.

↳ All 1's

255.255.255.255

Directed BC

↳ B/w different N/w

↳ NID + All 1's.

Eg. 10.15.20.26 ⇒ check class of N/w. (ie. A)

NID: 10.0.0.0

LBC: 255.255.255.255

DBC: 10.255.255.255.

for classless, find NID By ANDing (subnet mask & Host ID).

↓

check for NID.

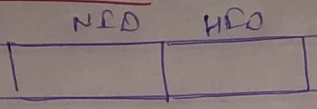
Sub:

→ If two NID matches, then check for longest subnet mask.

→ If nothing match then default.

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⇒ Classless Interdomain:



a.b.c.d /y

↓

Bits for NID.

eg. 20.10.50.100/20

NID = 20

HID = 32 - 20 = 12

IPs = 2^{12}

$2^{12} - 2$

Rules:

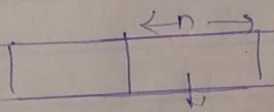
↳ All IPs address should be Continuous.

↳ Block Address size should be power of 2. [2^n].

↳ first IP Add. in the block should be evenly divisible by size of block.

[Reminder is Least signf bits]

eg. Block size 2^n



000 ... 0

Range

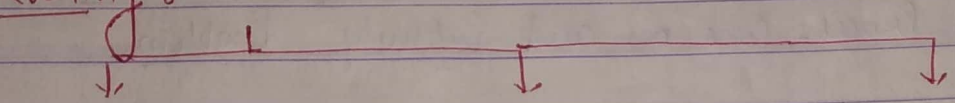
111 ... 1 Also.

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Routing

→ **Flooding:** ↳ send packet to all outgoing interface.
↳ No routing required.
↳ Highly reliable [one path may go down, it may go from other path]
↳ shortest path is guaranteed.
↳ Duplicate packets ⇒ High Traffic

→ **Routing:**



Static / Non-Adaptive

Dynamic / Adaptive
- DVR
- LSR.

Default Routing
All Routers are configured to send to single Router
↓
Stub.

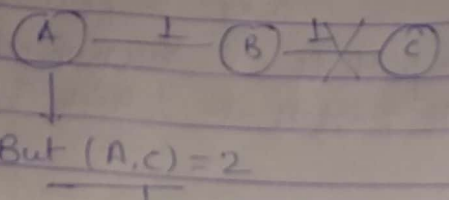
⇒ **Distance Vector Routing:**

~~↳~~ ↳ RIP [16 hops, Appl^{n} Layer protocol]
↳ OLD APRANET
↳ Updates on N/W are exchanged periodically (So, Bandwidth Waste)
↳ forward Complete table to neighbours so LOAD
↳ Routers always trust on routing inform^{n}. So, Rumers.
↳ Bellman-ford Basis

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- only local info present not Global I.for.
- loops may occur.
- Slower Compared to link state.
- Count-to-Infinity Problem.

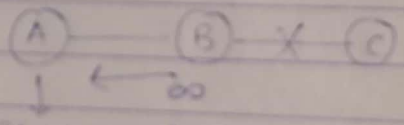
Count-to-Infinity Problem:



A	
A	0
B	1
C	2

B	
A	1
B	0
C	1

Soln: (1) Route Poisoning:



(2) Split Horizon:

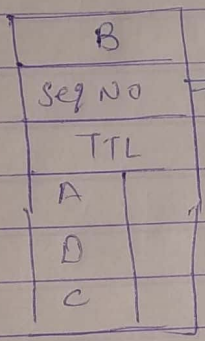
If the link b/w B and C goes down & B had received a route from A, B could end up using that route via A. A would send the packet back to B, creating a loop. B will ignore A info for C b/c B will never go to C by A.

(3) slowdown/Hold Down time:

Wait to wake up Again of Broked Conn. (wait for some Activity) from that node.

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⇒ Link state Routing :



→ Link state packet of B.

Sync the N/W.

→ High / fast Convergence.
→ Uses Dijkstra's Algo
→ No Count to Infinity Problm
→ Flooding [More Traffic]
→ Global knowledge.
→ OSPF (open source path first)

** → N/W Layer Protocol.

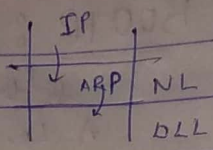
↳ OSPF :

↳ Interior Gateway Protocol
↳ Network Layer Protocol
↳ classless Protocol
↳ Both IPv4 & IPv6 routed.

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Protocols

ARP (Address Resolution Protocol):



↳ IP → MAC

↳ ARP would do create ia ARP request packet. only desired one will reply with it's MAC address

- ↳ ARP request is broadcasting
- ↳ ARP reply is unicasting
- ↳ ARP can send request with N/w only. (that's can request Router only if Host is out of Range).

↳ Host → Host (MAC Add)
Host → Router
Router → Router
Router → Host

RARP:

→ work at N/w layer.

↳ MAC → IP

- ↳ RARP message is broadcast and
- ↳ RARP reply is unicast.

↳ If RARP server is not present in same N/w then no reply. so, every N/w have RARP server.

↳ static mapping table only used.

↳ It is obsolete now.

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⇒ **BOOTP** : → Comes to overcome RARP.

* ↳ Work at application layer.

* ↳ Also used Static Mapping table.

↳ Relay Agent is used to avoid broadcasting and BOOTP server at every N/w.

↓

Single BOOTP.

↳ It knows the address of BOOTPs. It would send a unicast packet.

⇒ **DHCP (Dynamic Host Configuration Protocol)**:

↳ overcome BOOTP protocol by not using static mapping table.

- static [for servers, Webserver, Email, file server etc]
- Dynamic [Pool of IPs, only public IPs are present]

Practically, with routers only we have DHCP. So, No need of Relay Agents.

↳ Every N/w has Relay Agent.

↳ only one DHCP server is enough.

** ↳ Application Layer protocol [Uses TCP]

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⇒ ICMP (Internet Control Msg Protocol):

↳ Works at N/W Layer. → Unreliable

↳
- → Error Handling & feedback: [If packet corrupt then send ICMP, Not Mandatory]
- → Request & Reply

↳ Trace Route info can be found using ICMP.

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Transport Layer

⇒ Congestion Control:

Open Loop
(Prevent the Congestion)

During Comm^{n} setup, the transmission rate is determined.

"Once the connection is set-up, packets are sent and processed by the system without feedback info^{n} from either the N/w or destination."

- ↳ policing [Leaky Bucket & Token Bucket] (Traffic shaping)
- ↳ call Admission Control: All Resources got fixed.
  [QoS parameter will define arrival rate / Traffic rate etc in a N/w]

Close Loop / Feedback Based Control
(Remove the Congestion.

"feedback info is provided by N/w or destination"

- ↳ Senders can change forward speed dynamically.
- ↳ Back pass : Node to Node
- ↳ Choke packet : Direct to sender
- ↳ Explicit signal : To sender with msg.
- ↳ Implicit signal : source guess for congestion if no Ack reached.

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